Decision Trees for Customer Segmentation

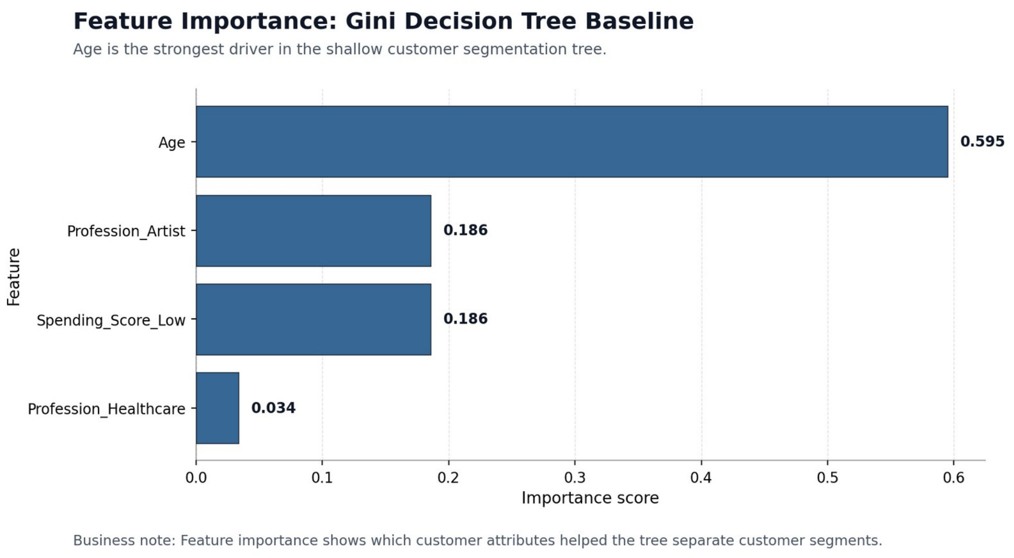
BI-ready executive presentation | Learning Day 7

CUSTOMER ANALYTICS

EXPLAINABLE ML

Executive headline

A shallow Gini Decision Tree is the recommended explainable baseline: 51.1% test accuracy, root split on Age, and clean business interpretation for customer segments.



Final deliverable: explainable, stakeholder-ready model story

Executive Summary

What the Decision Tree analysis tells the business

Gini Tree
Recommended model
Best explainability-performance balance

0.5105
Test accuracy
Baseline max_depth=3

Age
Top driver
Root feature at threshold 34

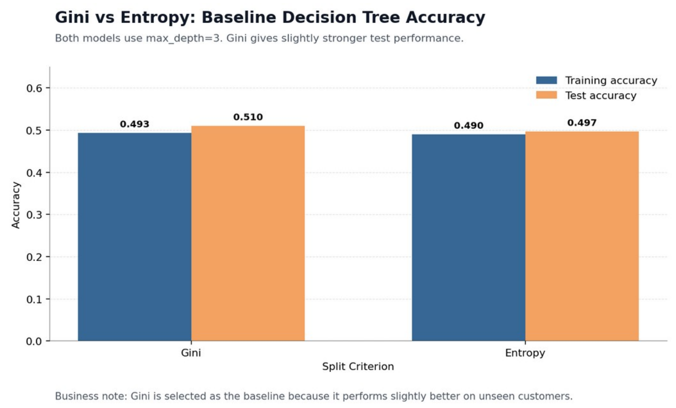
D
Best segment
Highest F1-score: 0.671

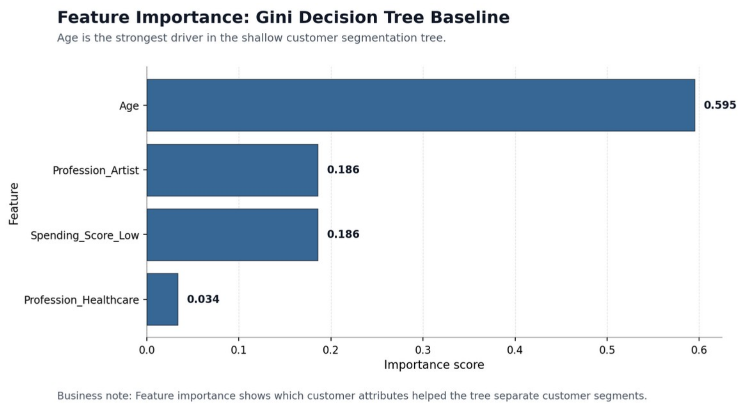
Business implication

The dataset is best served by a simple, explainable tree: younger, low-spending profiles strongly influence Segment D, while Segments A/B/C overlap more.

Modelling implication

The shallow tree generalizes without overfitting. Heavier trees memorize the training data; light pruning keeps accuracy stable while reducing complexity.





Dataset & Customer Segment Landscape

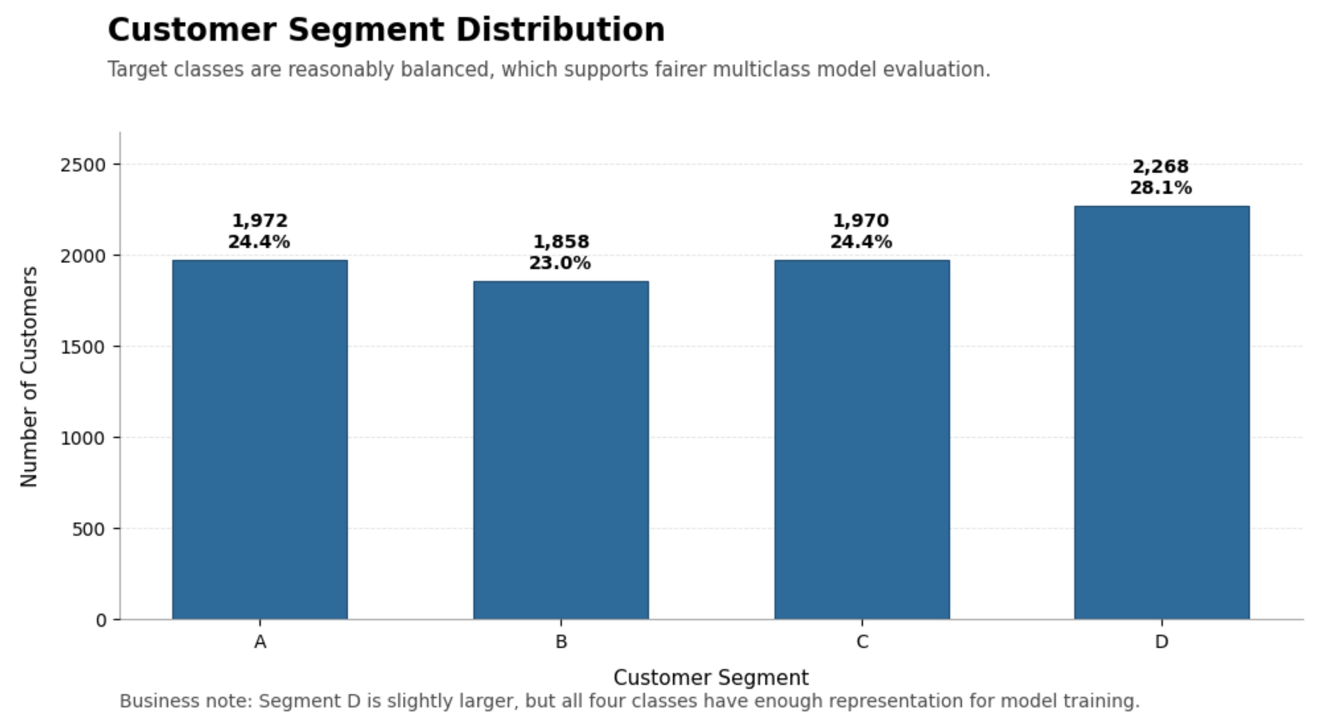
Kaggle customer segmentation dataset prepared for BI-style classification

Dataset snapshot

8,068 training rows, 11 original columns and 4 customer segments. Target classes are fairly balanced, which makes accuracy and segment-level evaluation meaningful.

Preprocessing

3 numerical features and 6 categorical features were transformed into 28 model-ready columns using imputation and one-hot encoding.



Target distribution is balanced enough for a fair multiclass baseline.

A	24.4%
B	23.0%
C	24.4%
D	28.1%

Modelling Workflow

From raw customer data to explainable BI insights

01	**Load & audit**	Rows, columns, duplicates and missing values
02	**Preprocess**	Impute missing values and one-hot encode categories
03	**Train baseline**	DecisionTreeClassifier with max_depth=3
04	**Evaluate**	Accuracy, classification report and confusion matrix
05	**Tune & explain**	Pruning, feature importance, oblique comparison and SHAP

Why Decision Trees?

Decision Trees are interpretable by design: rules can be visualized, root drivers can be explained, and business stakeholders can inspect how the model separates customer segments.

BI framing

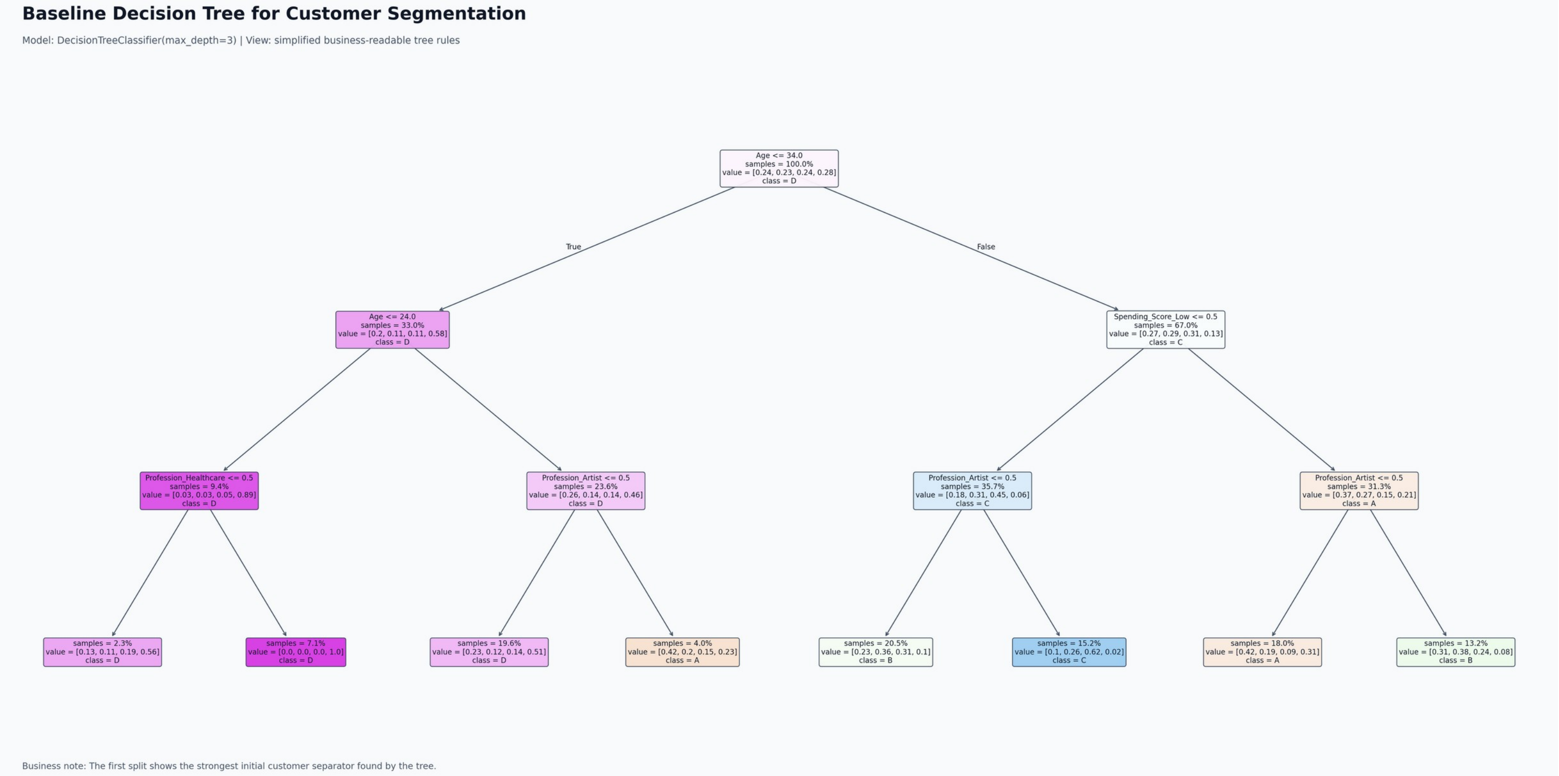
The project tells a full analytics story: data quality, modelling, validation, tuning, explainability and final recommendation.

Final output

A clear, explainable baseline model supported by SHAP and a documented MVD checklist.

Baseline Decision Tree

A shallow tree keeps the model explainable



Age <= 34

Root split

First and strongest separator

3

Depth

Interpretable by design

8

Leaves

Compact rule set

Key interpretation

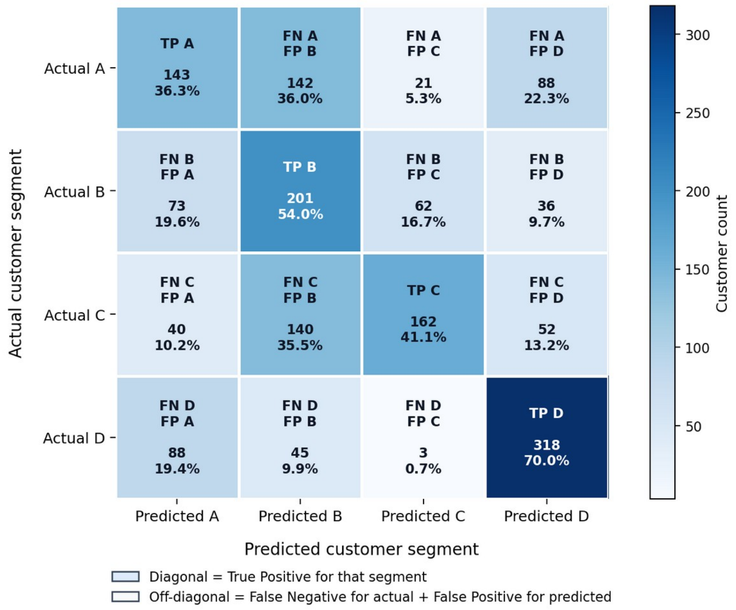
Age is the first question asked by the model. This suggests customer life stage is the strongest first-level segmentation driver.

Baseline Performance & Segment Confusion

Accuracy is useful, but the confusion matrix shows where segments overlap

Confusion Matrix: Gini Decision Tree Baseline

Multiclass customer segmentation evaluation | Test accuracy: 0.511



Business note: Segment D is classified best. Off-diagonal cells show where customer groups overlap and confuse the model.

Segment-level findings

Segment D is classified best. Segments A, B and C overlap more, especially A -> B and C -> B misclassifications.

Segment	Precision	Recall	F1
A	0.416	0.363	0.388
B	0.381	0.540	0.447
C	0.653	0.411	0.505
D	0.644	0.700	0.671

Business meaning

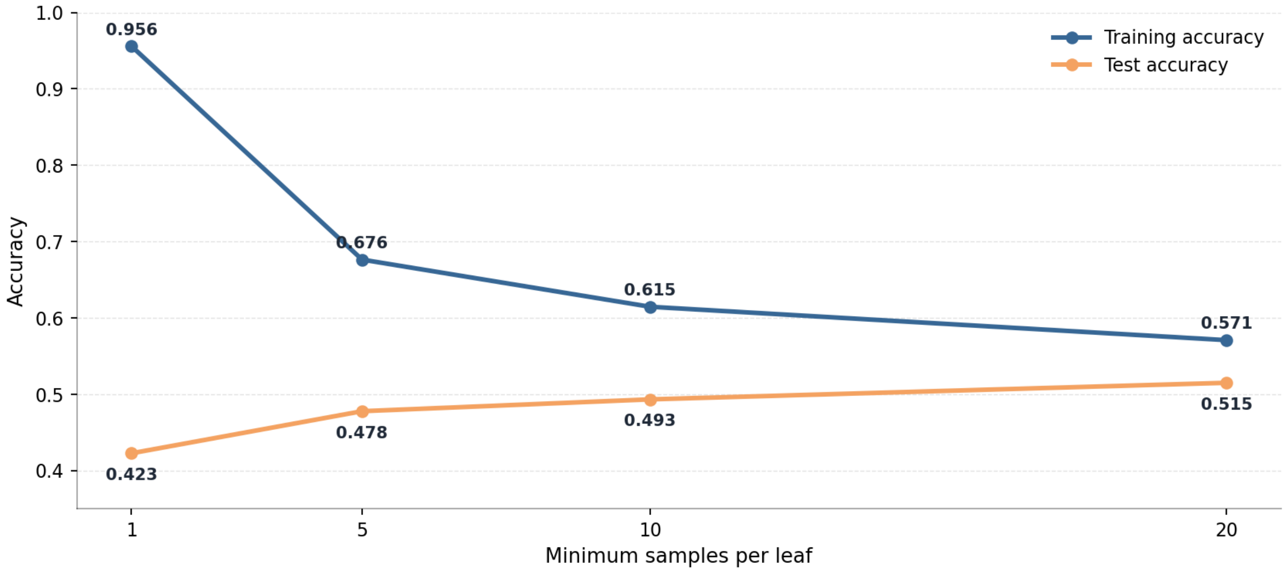
The model works best when customer profiles are distinct. Lower performance for A/B/C suggests these segments need richer features or stronger models.

Overfitting Control: min_samples_leaf

Larger leaves create broader, more stable customer rules

Effect of min_samples_leaf on Decision Tree Accuracy

Increasing the minimum leaf size reduces overfitting and improves test accuracy in this baseline experiment.



Business note: min_samples_leaf=20 gives the strongest test accuracy and the smallest train-test gap among these settings.

Pattern observed

min_samples_leaf=1 severely overfits: 95.6% train accuracy vs 42.3% test accuracy. Increasing the leaf size reduces complexity and improves generalization.

20

Best tested leaf size

Highest test accuracy in this set

0.5149

Test accuracy

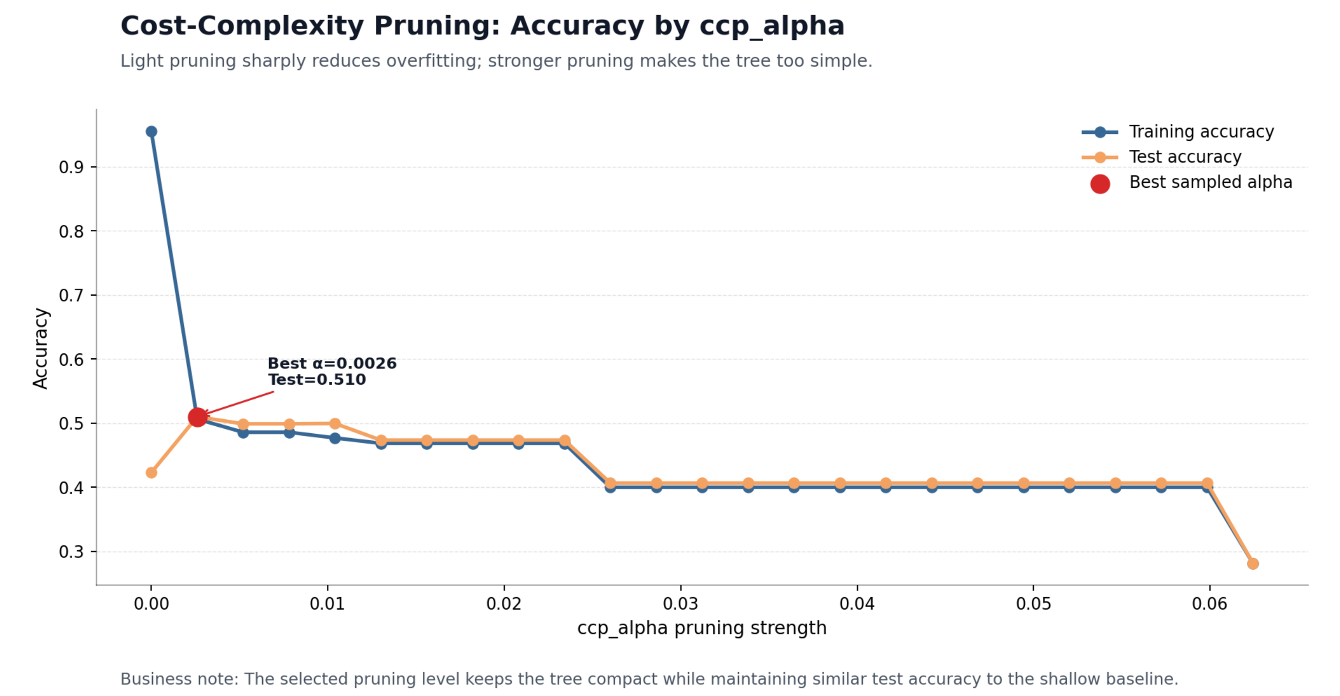
Strongest in leaf-size sweep

Conclusion

No underfitting appears within 1, 5, 10 and 20. The test score keeps improving while the train-test gap shrinks.

Cost-Complexity Pruning

Light pruning stabilizes the tree while keeping performance nearly unchanged



0.002601
Selected alpha
Best sampled pruning strength

4
Pruned depth
Compact model

0.5099
Test accuracy
Comparable to baseline

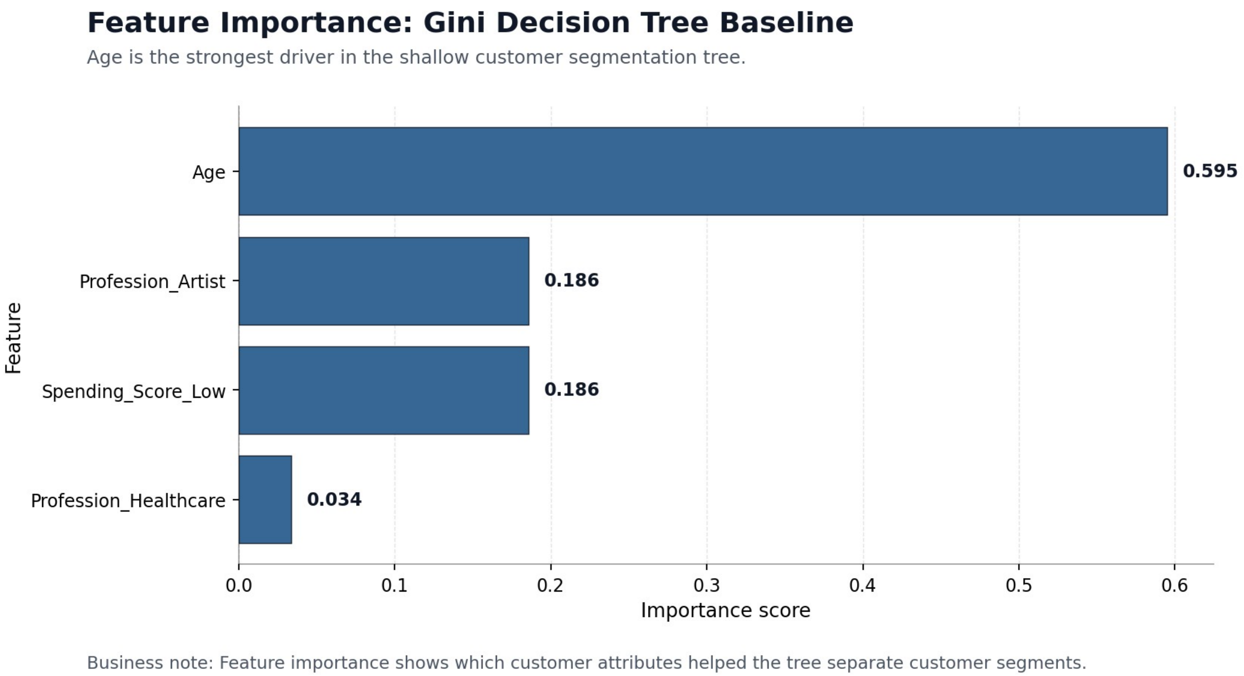
10
Leaves
Interpretable size

Interpretation

The unpruned tree memorizes training data. A small pruning penalty removes weak branches and keeps the tree stable.

Feature Importance: Business Drivers

The shallow tree relies on a small set of explainable variables



Top driver

Age accounts for ~59.5% of the tree importance and is also the root split. Customer life stage is therefore the strongest initial segmentation signal.

Secondary signals

Profession_Artist and Spending_Score_Low each contribute ~18.6%. These signals help the tree refine segment boundaries after the age split.

BI takeaway

For stakeholder reporting, focus first on age bands, spending score, and profession groups rather than all 28 encoded columns.

Expert Comparison: Standard vs Oblique Tree

More flexible splits did not outperform the simpler tree

Model	Split type	Depth	Leaves	Test acc.
Standard Decision Tree	One feature	3	8	0.5105
Custom Oblique Tree	Linear combo	3	8	0.4758

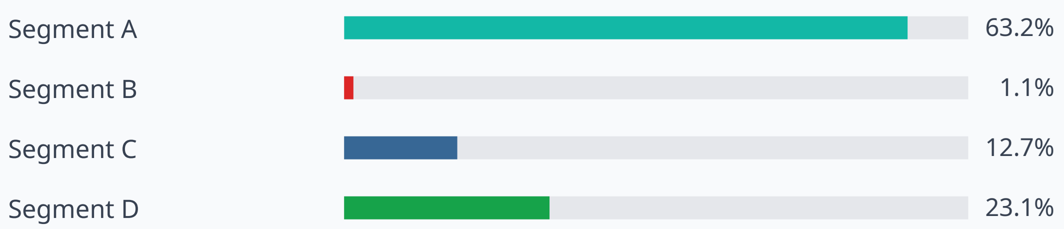
Why standard wins here

The customer segmentation rules appear well captured by simple feature splits. The oblique tree predicts all four classes, but its predictions are heavily biased toward Segment A.

Oblique prediction distribution

A: 63.2% | B: 1.1% | C: 12.7% | D: 23.1%. This imbalance explains why the oblique tree is less useful for this dataset.

Prediction distribution: Custom Oblique Tree

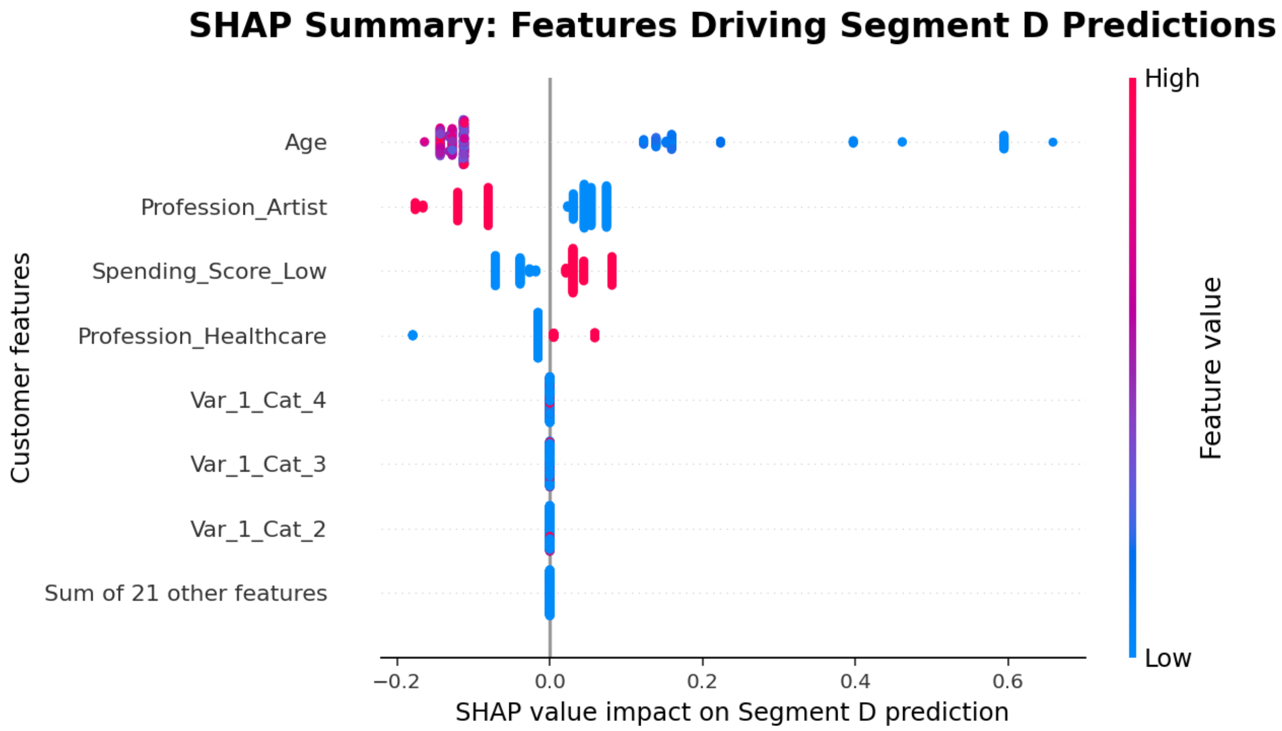


Recommendation

Use the standard Gini Decision Tree as the final explainable baseline. Keep the oblique tree as an expert-level comparison, not as the business recommendation.

SHAP Explainability: Segment D

Global explanation of what pushes predictions toward Segment D



How to read the plot

Right side = pushes toward Segment D. Left side = pushes away. Red means high feature value; blue means low feature value.

Interpretation

Lower Age pushes strongly toward Segment D. Low spending score also pushes toward D. Being an Artist tends to push away from D.

Var_1 note

Var_1 is an anonymized category from the dataset. Its business meaning is not provided, so it should be treated as an anonymous customer category.

Final Recommendation & MVD Coverage

The project meets the Decision Trees requirements and produces a clear BI story

Final model recommendation

Use the standard Gini Decision Tree as the explainable baseline: simple rules, readable visuals, stable test performance, and stakeholder-friendly interpretation.

Next modelling step

For production-level accuracy, compare against Random Forest, Gradient Boosting, XGBoost or LightGBM while keeping the Decision Tree as the explainability baseline.

MVD checklist

DecisionTreeClassifier(max_depth=3)	**Done**
Train/test accuracy + root feature	**Done**
Gini vs Entropy	**Done**
min_samples_leaf + ccp_alpha pruning	**Done**
Feature importance bar chart	**Done**
Standard vs Oblique comparison	**Done**
SHAP beeswarm + waterfall	**Done**

Final takeaway: keep the model simple when business explainability is the priority.